THE OTTER SIDE OF GROWING UP:

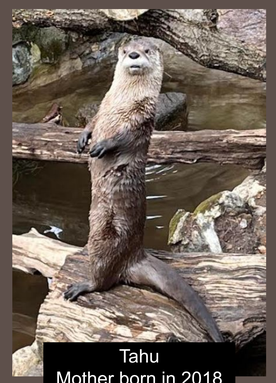
Exploring The Complex Social Dynamics Between a Mother-Daughter Pair of North American River Otters

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BACKGROUND

The North American River Otter (*Lontra canadensis*)is a largely solitary animal. In the wild, river otters have displayed levels of social plasticity, forming large groups depending on the environment[1]. However, this behavior is exhibited particularly among males and not in females. Previous research has indicated that river otters leave the family unit eight months after birth or once the next litter is born. While some of the reproductive and social behaviors of river otters have been documented, there is a significant gap in current knowledge. Thus, there is an urgent need to enhance the existing knowledge surrounding river otter behavior.



Tahu
Mother born in 2018



Flo
Daughter born in 2023

THE QUESTION

How does the sexual maturity of female river otters impact the social dynamics and interactions between mother (Tahu) and daughter (Flo) within the family unit?

WHAT WE DID

From January 2025 to April 2025, we conducted in-person observations of two North American River Otters, Flo and Tahu, at the Connecticut's Beardsley Zoo (CBZ), for one hour, three days a week. During these observations, we documented the incidences of common behaviors such as play, auto-grooming, allo-grooming, wandering, aggression, scent marking, and estrus behavior. Additionally, we recorded Flo and Tahu for thirty minutes once a week using a GoPro camera to measure the duration of Flo and Tahu were observed to be together and apart from each other.

SHIFT IN FLO'S BEHAVIOR AFTER SEXUAL MATURITY

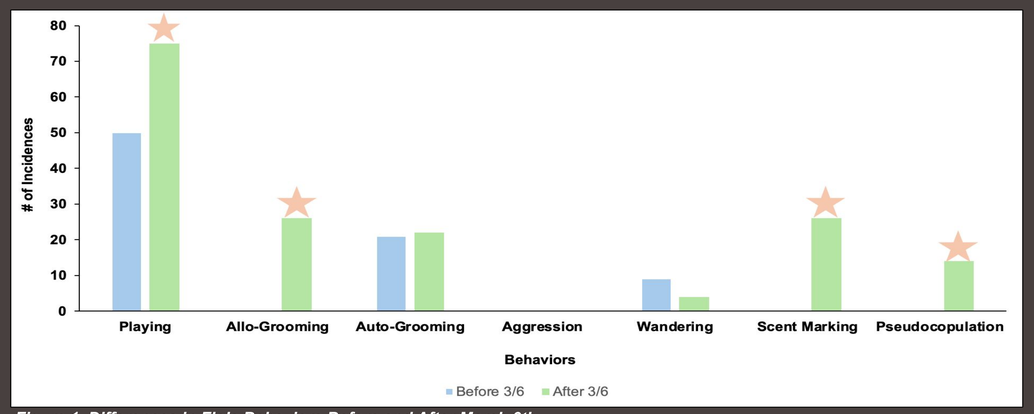


Figure 1. Differences in Flo's Behaviors Before and After March 6th
To determine a shift in Flo's behavior, we documented all incidences of Flo and Tahu's behavior between January to April 2025 with notable differences observed before and after March 6th, 2025, the date we believe Flo entered her first estrus period. We identified a significant increase in the number of incidences of playing, allo-grooming, scent marking, and estrus behaviors ($p < 0.05$; ★) after March 6th.

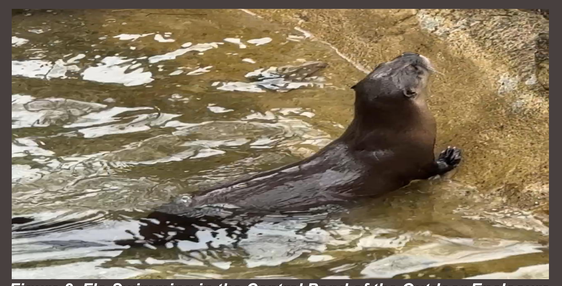


Figure 2. Flo Swimming in the Central Pond of the Outdoor Enclosure
To fully appreciate behaviors exhibited by Flo and Tahu, visual designations are key. The figure above shows Flo swimming in the central pond of the outdoor enclosure which was classified as "playing" behavior.

Classification of otter behavior can be a daunting and, somewhat, confusing task. Scan the QR code to the right to see all of Flo and Tahu's behaviors we examined throughout the semester!

SCAN ME

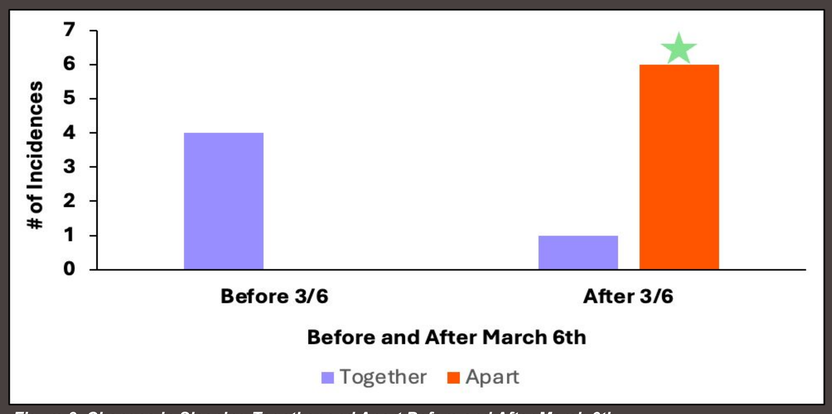


Figure 3. Changes in Sleeping Together and Apart Before and After March 6th
While evaluating shifts in Flo and Tahu's behavior throughout the duration of this study, we also documented the number of incidences when Flo and Tahu were observed to be sleeping together and apart from each other. We noticed a significant increase in the number of incidences of Flo and Tahu sleeping apart ($p < 0.05$; ★) after March 6th.

FLO DISPLAYS SIGNS OF SEXUAL MATURITY

At 2 years old, Flo is entering her first estrus period and there is an evident shift in her behavior which includes observed incidences of pseudo-copulatory behaviors (i.e. attempted mounting and vigorous play) towards Tahu. While observed in other mammals such as the Hanuman Langurs[2] and Alaska Fur Seals[3], no such behaviors have been documented in river otters, emphasizing the existing knowledge gap in river otter social and reproductive behaviors.

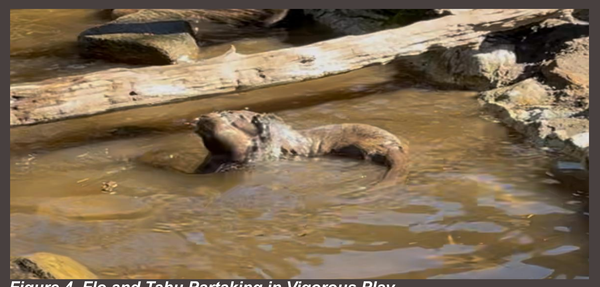


Figure 4. Flo and Tahu Partaking in Vigorous Play
To provide context regarding these incidences of behaviors correlated with sexual maturity, visual representations are required. The figure above shows Flo and Tahu engaging in vigorous play, a behavior indicative of copulation.

During our observations, we noted several incidences of Flo and Tahu engaging in pseudo-copulation such as attempted mounting and vigorous play. To visualize Flo and Tahu engaging in this behaviors, scan the QR code above!

WHAT'S NEXT?

We will continue our current in-person behavioral observations to document any potential increases in estrus behaviors such as scent marking, vigorous plays, and other activities related to copulation. Additionally, we will also continue to monitor for any shifts in their social dynamics and behavioral changes related to this change such as aggression and further independence observed by Flo. We hope that the findings of our study can be expanded to other AZA institutions to help formulate care plans for their housed population of river otters.

ACKNOWLEDGEMENTS

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